CS1910 Computer Science I • Expressions **Partner workbook**

Pair Problem Solving

In lecture with a partner • about 15 minutes

Name: _____ **Partner:** _____

This PDF can be filled in: click in an answer space to type, or print it and write. Draw diagrams with your PDF viewer's pen or markup tool, or on paper.

What this workbook practises

Two short problems that can both be solved with the arithmetic operators from lecture, *without* `if` statements. For each one, discuss with your partner first: what are you trying to achieve, and which operator helps?

1 Even or odd?

about 5 minutes

Problem 1

How can we determine whether a number is even or odd? *Difficulty: fairly straightforward*

1. Fill in the table. Which column tells you the answer?

n	`n // 2`	`n % 2`	**even or odd?**
6	3	0	even
7			
0			
15			
-4			

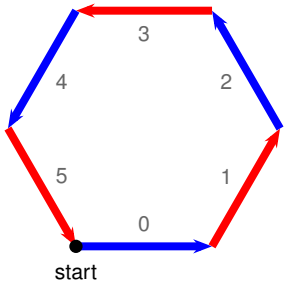
2. Write an expression that is `True` when `n` is even. Write one that gives `1` for odd numbers and `0` for even ones.

2 An alternating hexagon

about 10 minutes

Problem 2

How can we draw a hexagon with the turtle so that the sides alternate red and blue? *Difficulty: about the same*
Note: we are not using `if` statements yet.



Side numbers are the loop variable `i`. Turtle colours: `color("red")`, or `color(r, g, b)` with each of `r`, `g`, `b` from 0 to 255.

3. **Pattern recognition:** what repeats? How many times?

4. Describe (or write in Python) **two** different ways to draw the hexagon without an `if`.

Before you leave

about 1 minutes

- [ ] I can use `%` to test whether a number is even or odd.
- [ ] I can use an arithmetic expression to switch between two values without an `if`.